Unmixing detrital age distributions: a direct solution

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Abstract

Detrital geochronology can be used to quantify sediment budgets by estimating the proportions in which a set of known sources contribute to a sedimentary sink, such as a river downstream of a confluence. Existing 'top-down' unmixing algorithms solve this inverse problem by iterative search, either exhaustively testing all mixing proportions in 1% increments or sampling them at random. Their computational cost grows exponentially with the number of sources. They also rely on heuristic dissimilarity measures that lack a firm theoretical basis and can give inconsistent results depending on which measure is chosen. This paper presents a direct solution to the top-down unmixing problem. Because mixing of frequency distributions is a linear operation, the problem can be expressed in matrix form using empirical cumulative distribution functions and solved by least squares, subject to the constraint that the proportions sum to one. The solution is unique and is mathematically equivalent to minimising the Cramér–von Mises distance between the observed and predicted distributions, so no kernel smoothing or choice of objective function is needed. Its speed makes bootstrap resampling practical, which yields confidence intervals that capture sampling uncertainty in both sources and sink. Tests on simulated binary mixtures of normal distributions show that the method remains accurate even when the sources overlap strongly. Applied to a published detrital zircon U–Pb dataset from central California, the method gives confidence intervals that encompass the full range of results from the exhaustive search under five different dissimilarity functions. The Cramér–von Mises statistic also serves as a diagnostic for identifying sinks that the specified sources cannot adequately explain. As with any top-down approach, the results are only as reliable as the assumption that all relevant sources have been identified, and they describe the dated mineral phase rather than total sediment contributions. The algorithm is implemented in the `R` package `IsoplotR`.

1 Introduction

Consider the confluence of two rivers draining catchments with contrasting geological histories, and hence distinct detrital zircon U–Pb age distributions. Downstream of the confluence, the detrital zircon U–Pb age spectra form a mixture of the two input signals, with mixing proportions determined by the relative contributions of each branch to the zircon budget of the main river. If the zircon concentrations of both branches are known, these mixing proportions can be used to constrain the sediment budget of the river network. Solving this inverse problem is of considerable interest to sedimentary geologists, as it provides an opportunity to quantify sediment budgets using detrital geochronology. Sharman and Johnstone (2017) distinguish between top-down unmixing, in which the sources are known, and bottom-up unmixing, in which they are unknown. This paper focuses exclusively on the top-down problem because the bottom-up problem has no unique solution (Saylor et al., 2019).

Existing algorithms solve the top-down problem using iterative search strategies to find the mixing proportions that minimise the misfit between the observed and predicted age distributions. These algorithms are suboptimal in two respects. First, they are computationally inefficient. Sharman and Johnstone (2017) exhaustively search all possible mixing proportions in 1% increments, whereas Sundell and Saylor (2017)

use a Monte Carlo approach without a convergence criterion. Both approaches exhibit exponential time complexity: the computational cost of estimating the mixing proportions of a k-component mixture scales with the $(k-1)^{\text{th}}$ power of the number of possible proportions. Second, existing unmixing algorithms quantify the misfit between observed and predicted distributions using heuristic measures, such as 'likeness' and the 'cross-correlation of probability density plots'. These measures lack a firm theoretical foundation and have been shown to be problematic in limiting cases such as the application to high precision data (Vermeesch, 2018b).

This paper presents a direct solution to the unmixing problem that addresses both limitations. Section 2 shows that mixing frequency distributions is a linear problem that can be expressed in matrix form and solved using least squares estimation. This approach is mathematically equivalent to minimising the Cramér–von Mises distance between the observed and predicted distributions. Section 3 applies the new algorithm to simulated and published datasets to demonstrate its effectiveness. The examples also illustrate how the results can be used to identify problematic datasets for which the underlying model assumptions are violated. The new algorithm has been added to `IsoplotR` (Vermeesch, 2018a) for immediate use in detrital studies (Section 4).

2 A direct solution to the unmixing problem

The unmixing problem can be cast as a linear system of three matrices, using the following definitions:

k: the number of sediment sources (e.g. $k = 2$ in Figure 1);

W_j; the contribution of the jth source to the sink, where $W_k = 1 - \sum_{j=1}^{k-1} W_j$ (Figure 1d);

n: the sample size of the sink;

t_i: the ith sample (date) of the sink (with $1 \leq i \leq n$);

$X_{i,j}$: the cumulative probability of t_i under the empirical cumulative distribution function of source j (Figure 1e).

$Y_i = (2i-1)/2n$, which is essentially the empirical cumulative probability of t_i in the sink (Figure 1f).

Using these definitions, the mixing problem can be written as a linear matrix expression:

$$\begin{bmatrix} X_{1,1} & \dots & X_{1,j} & \dots & X_{1,k} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ X_{i,1} & \dots & X_{i,j} & \dots & X_{i,k} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ X_{n,1} & \dots & X_{n,j} & \dots & X_{n,k} \end{bmatrix} \begin{bmatrix} W_1 \\ \vdots \\ W_j \\ \vdots \\ W_k \end{bmatrix} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_i \\ \vdots \\ Y_n \end{bmatrix} \quad \text{or } X\,W = Y \tag{1}$$

It is worth noting that this matrix formulation is similar to the non-negative matrix factorisation approach introduced by Saylor et al. (2019) to solve the bottom-up unmixing problem. There are two key differences between their approach and the direct solution presented here. A first difference is that Saylor et al. (2019)'s X and Y are populated with density estimates, whereas the direct solution uses cumulative probabilities. This avoids the need for kernel smoothing. The second difference is that both X and W are unknown in Saylor et al. (2019)'s bottom-up formulation, whereas X is known and only W needs to be estimated in the top-down problem. Consequently, Equation 1 has a unique solution, obtained by minimising the sum of squared residuals between Y and XW, subject to the constraint that $\sum_{j=1}^{k} W_j = 1$. Several methods can be used to solve Equation 1, including Lagrange multipliers and non-negative least squares algorithms (Lawson and Hanson, 1995). Unlike non-negative matrix factorisation, this least squares problem has a unique solution.

Both sides of Equation 1 represent cumulative distributions. The sum of the squared differences between two such distributions is closely related to the Cramér–von Mises (CvM) criterion, a well-established statistical measure:

$$\mathrm{CvM} = \frac{1}{12n} + \|X\,\hat{W} - Y\|^2 \tag{2}$$

where $\hat{W}$ denotes the least squares solution. The Cramér–von Mises criterion does not satisfy the triangle inequality and is therefore not a proper metric. This limits its suitability for applications such as multidimensional scaling (Vermeesch, 2018b). For the purpose of unmixing, however, the Cramér–von Mises criterion is adequate.

The mixing proportions W_j are constrained to lie between 0 and 1. In the vicinity of either boundary, their uncertainty distributions become skewed and strongly correlated. Because the direct solution is computationally efficient, the method is well suited to bootstrap resampling, which captures sampling uncertainties in both the sources and the sink. Section 3 demonstrates the effectiveness of this approach using a simulated example.

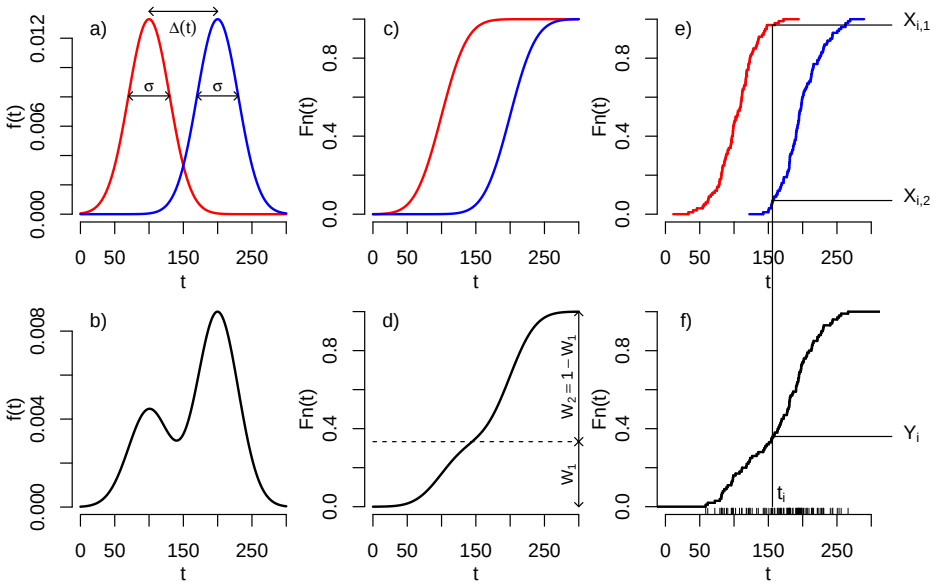


Figure 1: Schematic overview of the mixing problem using a bivariate mixture of normal distributions as an example. a) density functions of two synthetic source populations, and b) a 1/2-mixture of these sources (i.e., $W_1 = 1/3$ and $W_2 = 2/3$). c) and d) are the corresponding cumulative distributions. e) and f) are empirical cumulative distribution functions of random samples of 100 values from each of the three populations. $X_{1,i}$, $X_{2,i}$ and Y_i are the cumulative probabilities under the source and sink distributions for the ith measurement. These values are used to populate Equation 1.

3 Examples

As a first example, consider a binary mixture of two normal distributions with the same standard deviation σ (Figure 1a). Let $\Delta(t)$ denote the lateral separation between the two source distributions. When $\Delta(t)$ is much larger than σ, the sink distribution is bimodal, making the mixing proportions easy to recover. As $\Delta(t)$ decreases, the sink becomes unimodal, making the deconvolution problem more difficult. Figure 2 shows the bootstrapped 95% confidence intervals for W_1 as $\Delta(t)$ varies from 0 to 3 standard deviations. As expected, the uncertainties decrease with increasing separation. Nevertheless, the direct solution remains accurate even when $\Delta(t)$ is small.

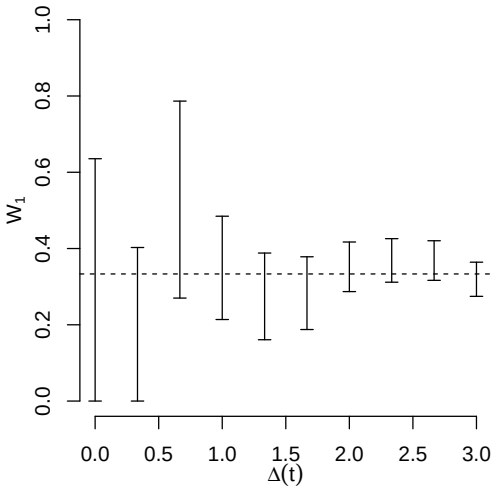


Figure 2: Bootstrapped 95% confidence intervals for the mixing proportion of synthetic samples formed by 1/2 mixtures of two normal distributions with offset $\Delta(t)$, which is expressed as a number of standard deviations. The dashed line shows the true value for the mixing proportion W_1.

Figure 3 applies the direct solution to the detrital zircon U–Pb dataset of Sickmann et al. (2016), which was used as a case study in the unmixing study of Sharman and Johnstone (2017). The dataset comprises seventeen sand samples from central California (USA): five river samples, four beach samples and eight submarine canyon samples. Sharman and Johnstone (2017) partitioned these data into three sources (P1, P2 and P3) and twelve sinks. We retain the same partitioning to facilitate comparison between the direct solution and the exhaustive search algorithm of Sharman and Johnstone (2017).

Figure 3 consists of three parts. The top panel shows the empirical cumulative distributions of the three sources. P1 and P2 have broad U–Pb age distributions, whereas P3 is more concentrated in the 80–100 Ma interval. The left-hand panels show the observed (Y, red) and fitted ($X\hat{W}$, blue) distributions for each of the twelve sinks. The goodness of fit is quantified using the Cramér–von Mises statistic (CvM), with higher values indicating a poorer fit. Sample GSB has the highest CvM value in Figure 3. Closer inspection reveals that its DZ dates span a much narrower range than those of any of the three sources, explaining the poor fit. Thus, the CvM statistic provides a useful diagnostic for identifying potential problems with the unmixing solution. Finally, the lower-right panel shows the estimated mixing proportions for all twelve sinks.

Before comparing the direct solution with the exhaustive search algorithm of Sharman and Johnstone (2017), it is worth noting that the results of the latter depend on the objective function used for optimisation. Sharman and Johnstone (2017) explored five dissimilarity functions (likeness, similarity, cross-correlation, Kolmogorov–Smirnov and Kuiper), each yielding different results. They illustrate this sensitivity using sample MAN, for which the estimated mixing proportions span the following ranges across the five functions: $66\% \leq W_1 \leq 86\%$, $3\% \leq W_2 \leq 15\%$ and $0\% \leq W_3 \leq 31\%$. By comparison, the direct solution for MAN yields $\{W_1 = 83\%, W_2 = 5\%, W_3 = 12\%\}$, with 95% confidence intervals of $66\% \leq W_1 \leq 90\%$, $0\% \leq W_2 \leq 20\%$ and $0\% \leq W_3 \leq 31\%$. These confidence intervals encompass the full range of solutions obtained by Sharman and Johnstone (2017), while avoiding the ambiguity introduced by the choice of dissimilarity function.

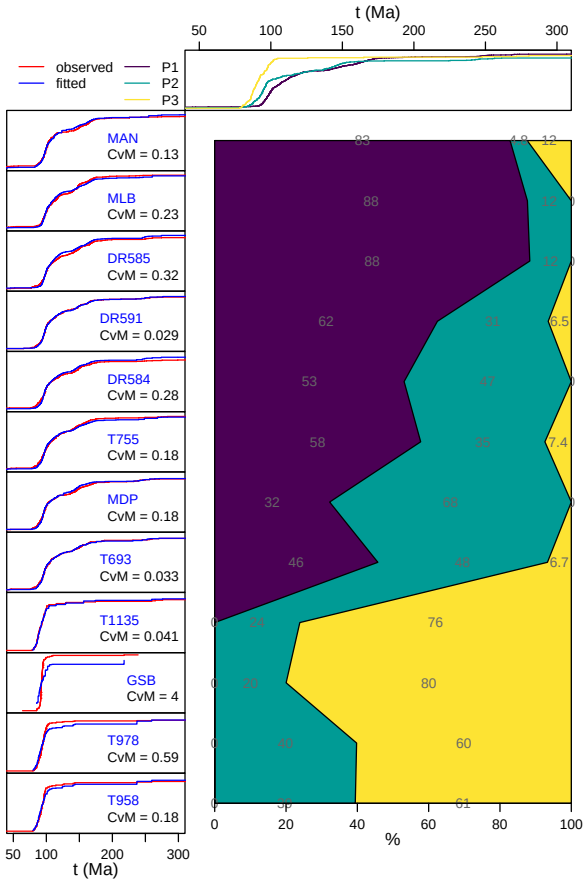


Figure 3: Unmixing the detrital zircon U–Pb dataset of Sickmann et al. (2016). The top and left margins show the three sources (P1, P2 and P3) are twelve sinks, respectively. 'CvM' is the Cramér–von Mises misfit of the optimal fit. The main panel shows the mixing proportions of the different samples.

4 Implementation in IsoplotR

The new algorithm was implemented in the `R` package `IsoplotR` (version 7.1 and higher) via a new `unmix` function. It uses a modified least squares algorithm that enforces a unit sum of the mixing proportions with a Lagrange multiplier. If the solution returns any value outside the unit interval, a fallback solution uses the non-negative least squares algorithm of Lawson and Hanson (1995) with a penalty applied to the final row to enforce the unit sum of the weights. The `unmix` function can either be called from the `R` console, or from `IsoplotR`'s graphical user interface (GUI), which is available online and offline Vermeesch (2018a). In the GUI, the unmixing functions can be accessed by selecting 'detritals' and 'unmix' as top-level options. At the console:

```
library(IsoplotR)
DZ <- read.data("Sickmann.csv",method="detritals")
unmix(DZ,source_names=c("P1","P2","P3"))
```

To obtain bootstrapped confidence intervals:

```
fit <- unmix(DZ,source_names=c("P1","P2","P3"),boot=TRUE,plot=FALSE)
```

which returns a list containing matrices called `W`, `ll` and `ul`, storing the mixing proportions and the lower and upper limits of their confidence intervals, respectively.

5 Conclusions

This paper introduced a direct solution to the top-down sediment unmixing problem, which had previously been addressed using exhaustive and random search algorithms. Whereas these numerical approaches become progressively slower as the number of sources increases, the direct solution can be computed almost instantaneously on any modern computer. The algorithm can accommodate any number of sources, provided that this number does not exceed the number of dates in the sink.

Beyond its obvious speed advantage, the direct solution eliminates the ambiguity associated with choosing between different objective functions in iterative approaches, some of which lack a firm statistical basis. Instead, the new algorithm uses least squares to minimise a single, well-established objective function: the Cramér–von Mises (CvM) statistic. When applied to multiple sinks, the CvM statistic can help identify problematic samples that cannot be adequately reproduced by the specified sources. In principle, it can also be used to formally test whether a sink represents a mixture of those sources. However, such tests have limited practical value because their outcomes depend strongly on sample size. Given a sufficiently large sample, no geological sink will be an exact mixture of a finite number of sources. The CvM statistic is therefore better used as a relative measure of goodness of fit than as a basis for calculating p-values (Vermeesch, 2018b).

As with any top-down unmixing model, the reliability of the results depends on the validity of the input. Whether the estimated mixing proportions reflect geological reality or an analytical artefact depends on the assumption that all relevant sources have been identified. Moreover, even if this assumption holds, the estimated weights quantify only the contributions of the datable mineral, not the total sediment. Converting these proportions into sediment budgets requires estimates of mineral fertility (Malusà et al., 2016), which are not always available in real-world studies.

Acknowledgments

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Supplementary Information

All the code and data to generate the figures in this paper are available online at `https://github.com/pvermees/supplements`.

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